Bus Servo Introduction

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Bus servos are the derivative of digital servos, but differ from traditional PWM digital servos. They communicate via asynchronous serial bus, allowing for control through the sending and receiving of instruction packets. This closed-loop control method allows bus servos to be connected in series, simplifying wiring and reducing the use of serial ports.

Therefore, when connecting bus servos, just like giving names to people, each servo must be assigned a unique ID beforehand, otherwise they cannot be distinguished during control. This allows commands such as “Rotate servo ID1 by 30 degrees” and “Rotate servo ID2 by 40 degrees” to be sent when communicating with bus servos.

Bus servos adopt high-precision potentiometers internally, and compared to PWM servos, they not only have the ability to provide feedback on information such as position, temperature, and voltage, but also have good precision and linearity, resulting in more stable robot operation and longer servo lifespan.

Additionally, before assembling the servo, it is necessary to perform a neutral operation to set the initial position, which is used as the "zero point" for positive and negative angle rotation. Therefore, the servo must be adjusted to the neutral position before installation onto the servo horn.

When the servo motor rotates, it moves the potentiometer to rotate. The software assumes this neutral position is the "zero point". Otherwise, the potentiometer might enter a "blind spot", causing the entire component to malfunction. This can result in the servo being unable to reach the specified angle or the action group corresponding to the angle being inconsistent during robot operation.